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TAJ Industries

B/12-C, S.I.T.E.
75700 Karachi
Sindh Pakistan

Latitude: [24.910152], longitude: [67.012005]
Date of inspection: 14-Apr-2026

Electrical Safety Inspection Report

Observations & Actions



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Pakistan Accord Initial Safety Inspection Report

About this report:

This report has been prepared by the Health and Safety Accord Pvt Ltd after visiting the factory on 14-04-2026 as part of the inspection and remediation program of the Pakistan Accord on Health and Safety in the Textile and Garment Industry. The purpose of the report is to provide key findings and identify related safety improvements needed at the inspected factories. This inspection report may not identify minor deficiencies, which may be further addressed as part of follow-up inspections.

In addition, while this round of initial inspections covers fire, electrical and structural safety, the Pakistan Accord intends to include inspections on boiler safety and hazardous material in later rounds.

This report will serve to facilitate discussions between the Pakistan Accord, the factories and the responsible brands that source from this facility about the remediation requirements. An implementation schedule shall be developed by the factory in the form of a Corrective Action Plan (CAP) to remediate each of the findings identified in the inspection report, including timelines and cost estimates, for approval by the Chief Safety Officer (CSO).

Applied Standards:

This safety assessment was conducted by the Pakistan Accord with reference to the Pakistan Accord Building Standard.

Contents

1. Executive Summary 5
2. Building Extents 6
   - 2.1 Building Use 8
   - 2.2 Utility Source 9
   - 2.3 Backup source 10
3. Documentation 11
4. Electrical Safety Measures 12
5. Diagrams & Drawings 13
   - 5.1 Single Line Diagram (SLD) 13
   - 5.2 Electrical layout drawing 15
6. DBs & Elec Protection Systems 16
   - 6.1 Circuit breakers 16
   - 6.2 Hot Spots 18
   - 6.3 Earthing system 19
   - 6.4 Panel/cabling/busbar 22
   - 6.5 Loop Connection 24
   - 6.6 Multiple Connections 26
   - 6.7 Openings 28
   - 6.8 Adequate accessibility 30
   - 6.9 Lint & dust inside panel board 32
   - 6.10 Lightning Protection System (LPS) 34
7. Fire Alarm & Detection System (FADS) 35

7.1 Non-functional or incomplete 35
8. Emergency Lighting 37
8.1 Provided but non-functional 37
8.2 Provided but insufficient 39
9. Generator, Transformer & Sub-Station Room 43
9.1 Battery 43
10. Cabling & Wiring 45
10.1 Cable support and protection 45
10.2 Splicing of cables 48
10.3 Colour coding 51
10.4 Fixtures 53
11. Assessment Limitations and Assumptions 55

1. Executive Summary

On 14-Apr-2026, Muhammad Asif and Zeeshan Haider on behalf of the Pakistan Accord carried out an initial electrical inspection for TAJ Industries, a factory at the address and coordinates given on the cover page of this report. The inspection was carried out in accordance with the Pakistan Accord Building standards.

We met with factory management representative Mr. Zeeshan (HR and Compliance Manager) and other staff members of TAJ Industries. As the assessment was being carried out as part of an overall Structural, Fire and Electrical initial inspection, other representatives from Pakistan Accord were present.

The factory operates in one shift for all departments. Approximately 250 workers operate in the factory with 05 workers being females.

The factory comprises of 01 main building owned by TAJ Industries. During the inspection, the buildings highlighted in Figure 1.1 were surveyed, with the site boundary indicated in red:
Main Building: Seven storey (Ground + Mezzanine + 5) building is used for compressor and livestock storage (Roof), warehouse (5th floor), cut to pack (4th floor), stitching, packing and checking (3rd floor), stitching, packing and checking (2nd floor), offices, prayer area, cutting, bundling and sampling (1st floor), offices, accessories store, parking, canteen and record room (mezzanine) and fabric inspection, warehouse, parking, genset room and canteen (Ground floor), constructed in 1999 and purchased in 2012.

A portion at mezzanine and ground floor level is rented by Allied bank Limited, Access to these areas was restricted; detailed data recording was not permitted, and only a brief walkthrough by a single person was allowed. However, factory and bank has a separate utility connection.

A summary of main concerns identified on-site during the assessment (non-exhaustive) that should be actioned are as follows:

Fire alarm system was found non-compliant, non-listed, having inadequate detection coverage issues, pose life safety risk in the facility.
Earthing system was found inadequate at different locations which pose life safety risk due to electric shock.

Our Limitations and Assumptions are also noted at the end of this report.

For detailed analysis of the above, please refer to the full inspection report below, which further details each issue, and also includes photographs. Further actions with associated priorities and timeframes are also detailed with each issue. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

2. Building Extents



Figure 1.1. Site plan (Satellite view) – Red dashed line indicates site boundary, highlighted area (blue) indicates building inspected

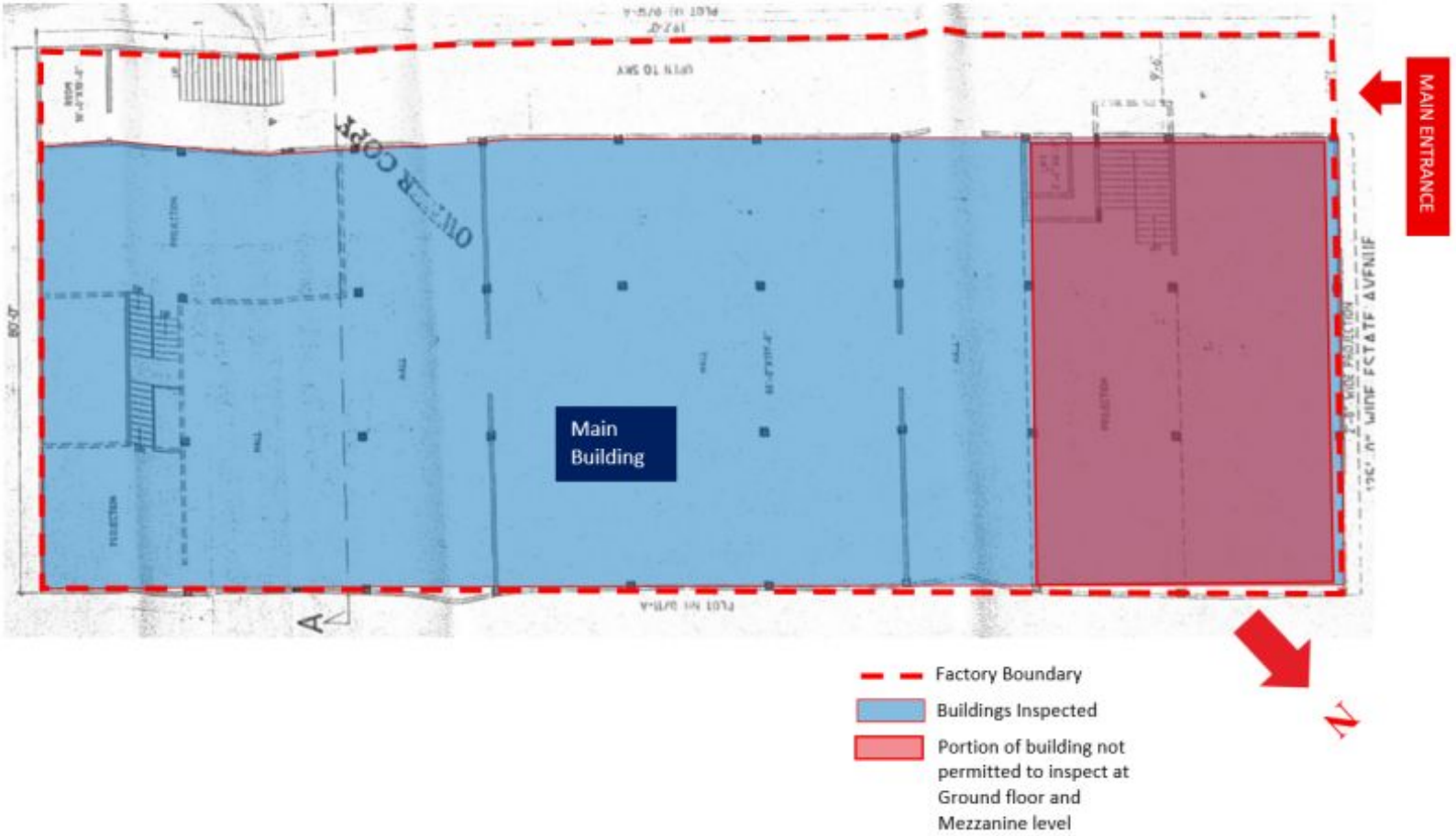


Figure 1.2. Site layout plan - Red dashed line indicates site boundary, highlighted area (blue) indicates buildings inspected and (red) indicates portion not allowed to inspect.

2.1 Building Use

Main Building: This facility consists of a seven storey (Ground + Mezzanine + 5) building. Details of the building are given below:

L00-Ground floor: used for fabric inspection, warehouse, parking, generator area and canteen.
M01-Mezzanine floor: used for parking, canteen, record room, offices and accessories store.
L01-First floor: used for offices, prayer area, cutting, bundling and sampling area.
L02-Second floor: used for stitching, packing and checking area.
L03-Third floor: used for stitching, packing and checking area.
L04- Fourth floor: used for cut to pack.
L05-Fifth floor: used for warehouse.
L99-Roof: used for compressor and livestock.

2.2 Utility Source

The facility has a low voltage electrical supply connection from a utility company K-Electric to the LT panels. From those LT panels, power is distributed on different floors through each floor distribution board.

2.3 Backup source

This facility has diesel generators of rating 150 and 250 kVA respectively, used as a backup supply arrangement. Moreover, fire alarm system and emergency lighting system have built-in battery backup.

3. Documentation

This table shows the details about the documentation provided by the factory.

S.NO	Description	Available (Y/N)	Created Date	Status / Remarks
1	Electrical As-Built Layout Drawing	No		
2	Single Line Diagrams	No		
3	Fire Alarm Drawing	No		
4	Earth Test Report	No		
5	Thermographic Test Report	No		
6	Insulation Resistance Test Report	No		
7	LPS Risk Assessments	No		
8	Periodic Maintenance Record	No		
9	Transformer Oil Testing Report	N/A		

4. Electrical Safety Measures

During our electrical site inspection of TAJ Industries, a number of electrical issues were identified.

It is essential to address these electrical safety issues promptly to enhance safety and compliance within TAJ Industries. Each issue should be carefully evaluated, and the recommendations provided in this report should be followed diligently to rectify these concerns. The safety and well-being of the factory's employees depend on these corrective actions. Where issues are identified in the report, they are grouped by topic, not in order of importance. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

Recommendations in this report are prioritized as follows:

Immediate: These recommendations address safety and compliance concerns that require immediate attention.

Medium term (within approx. 6-weeks): Recommendations of high importance but needing slightly more time to implement.

Longer term (within approx. 6-months): Important recommendations that require a longer period to fully implement or that enhance long-term safety and efficiency.

All survey, testing, and design activities specified in this report shall be performed by a qualified electrical engineer. The recommended installation and modification tasks should be executed by a licensed professional electrician. Before commencing any installation and/or modification works, the proposed design must be submitted to the Accord for approval.

Upon completion of the works, the necessary evidence required to verify that the tasks have been completed to the agreed Standards will be determined during the development of the Corrective Action Plan (CAP). Our Limitations and Assumptions are also noted at the end of this report.

5. Diagrams & Drawings

5.1 Single Line Diagram (SLD)

This section refers to the entire facility.

It was found that the electrical single line diagrams (SLDs) for the factory were missing.

This deficiency was noted in all electrical rooms and areas where SLDs should be prominently displayed.

The factory lacks documented diagrams that illustrate the electrical distribution system, including the connection and configuration of all electrical components.

Single line diagrams are crucial for understanding the electrical distribution network and for troubleshooting issues effectively. Without these diagrams, maintenance personnel and emergency responders lack critical information necessary for isolating faults and managing power during emergencies. This can lead to delays in addressing electrical issues, increasing the risk of electrical fires and potential harm to personnel. Moreover, the lack of SLDs complicates routine maintenance and inspection processes, potentially resulting in undetected electrical faults and system inefficiencies.F-202604-065761

Timeline	Remediation Plan
Medium Term CAP-0078189	Develop comprehensive electrical single line diagrams for the entire facility in accordance with the requirements outlined in Section 514.9.1 of BS7671 18th Edition + A2 (2022). Ensure all components and connections are accurately represented. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0078190	Implement a regular review and update process for the single line diagrams to reflect any changes in the electrical system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the diagrams.

5.2 Electrical layout drawing

This section refers to the entire facility.

During the survey, it was noted that the factory lacked electrical layout drawings - including small power layouts, lighting layouts, and containment layouts.

This absence was consistent across all areas where such documentation should be available.

These drawings are vital for understanding the placement and connections of electrical systems, which is crucial for identifying and mitigating potential hazards. In emergencies, the lack of detailed layouts can delay the identification and isolation of electrical faults, increasing the risk of fire and electrical accidents. Additionally, without proper layout drawings, routine inspections and maintenance may overlook critical issues, potentially leading to system failures and safety breaches.F-202604-065763

Timeline	Remediation Plan
Medium Term CAP-0078193	Develop comprehensive electrical layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). To avoid drawings becoming overcrowded, separate drawings shall be prepared for each electrical system. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0078194	Implement a regular review and update process for the electrical layout drawings to reflect any changes in the system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the layouts.

6. DBs & Elec Protection Systems

6.1 Circuit breakers

This section refers to the entire facility.

During the survey the circuit breakers installed throughout the facility inside the electrical panels and distribution boards were inspected.

It was observed that multiple circuit breakers had ratings that were not appropriate for the connected cable sizes.
1. 25 A circuit breaker was installed with 0.75 sq.mm cable in grey area at ground floor.
2. 150 A circuit breaker was installed with 6 sq.mm cable in grey area at ground floor.

Specifically, some circuit breakers had higher ratings than the cables could safely handle, which poses a significant risk of overheating and potential fire hazards. Conversely, other circuit breakers had lower ratings than necessary, resulting in frequent and unnecessary tripping. To ensure both safety and operational efficiency, corrective action is required to align the circuit breaker ratings with the cable capacities. This adjustment will mitigate the risks associated with overheating and improve the reliability of the electrical system. F-202604-065449

Timeline	Remediation Plan
Medium Term CAP-0077453	Evaluate the breaker ratings and cable sizes for all circuits in the facility according to Section 433.1.1 and Appendix 4 of BS 7671 18th Edition + A2 (2022). Ensure that the breaker rating does not exceed the cable's current-carrying capacity. If the breaker has adjustable settings, they must be configured to allow current within the cable's ampacity after applying derating factors.



Figure 2.1 - Undersized cables with oversized circuit breakers were found in LT room (150A breaker with 6 sq.mm cable)



Figure 2.2 - Undersized cables with oversized circuit breakers were found in LT room (150A breaker with 6 sq.mm cable)



Figure 2.3 - Undersized cables with oversized circuit breakers were found in LT room (25A breaker with 0.75 sq.mm cable)

6.2 Hot Spots

This relates to the entire facility.

During the inspection, Thermographic surveys were conducted on the electrical boards.

Thermographic inspection report was not found in the facility.

Circuit breakers are typically designed to operate safely within an ambient temperature range of -20°C to 40°C. Commonly used PVC insulated cables typically have a maximum operating temperature of 70°C. Prolonged exposure to temperatures above these can degrade the insulation of the breaker as well as the cable and result in reduced performance.

Hot spots in electrical systems pose significant life safety risks due to several factors. Overheated components can ignite surrounding materials, leading to electrical fires. This is particularly dangerous in densely packed panels and distribution boards where flammable materials may be present. Hot spots can degrade insulation on wires and components, increasing the risk of electrical shock to personnel who might come into contact with exposed conductive parts. Overheating can contribute to conditions that might trigger an arc flash, a dangerous high-temperature event that can cause severe injuries or fatalities to personnel working near the affected equipment. F-202604-065439

Timeline	Remediation Plan
Medium Term CAP-0077724	Regular thermographic inspections should be incorporated into the maintenance schedule to proactively manage and mitigate the risks associated with overheating after every 6 months.

6.3 Earthing system

This relates to the entire facility.

During the inspection of the facility's electrical panels, distribution boards, and machinery, several observations with the earthing system were identified.

i. The earthing cables were not appropriately sized according to the relevant earthing standards. This was observed at panel near main gate such as 35 sq.mm cable was installed with 95 sq.mm phase cables.
ii. The earthing system was absent in multiple locations such main earth bar was found with no connected cables. Several distribution boards such main panel near main gate and grey area DB lacked earthing cables, and many electrical machines did not have proper earth connections.
iii. Equipotential bonding for the cable trays and metallic panel covers was missing.
iv. The earth pits were inaccessible and unsuitably located. Moreover earthing design drawings were found missing.

The lack of proper earthing system throughout the facility poses substantial safety risks, such as electric shocks, fires, and equipment damage. Implementing appropriate earthing measures is crucial to prevent hazardous voltage levels and ensure operational safety. F-202604-065440

Timeline	Remediation Plan
Medium Term CAP-0077742	i. Replace all earthing cables with appropriately sized cables as per the requirements of BS 7671 18th Edition + A2 (2022) Appendix 4, Section 5. Verify that all earthing conductors meet the minimum cross-sectional area specified.
Medium Term CAP-0077743	ii. Install earthing systems where absent, ensuring compliance with BS 7671 18th Edition + A2 (2022) Chapter 54. Fit all electrical boards with the required earthing cables. Ensure that earthing conductors are properly connected and comply with BS 7671 18th Edition + A2 (2022) Section 543.1.
Medium Term CAP-0078180	iii. Implement equipotential bonding for all cable trays to comply with BS 7671 Regulation 411.3.1.2. Ensure all exposed conductive parts are bonded to the main earthing terminal.
Medium Term CAP-0078181	iv. Relocate and redesign earth pits to make them easily accessible for maintenance and inspection, complying with BS 7671 18th Edition + A2 (2022) Chapter 54. Develop earthing design drawings and submit to Accord for its review.

Longer Term CAP-0078182	Establish a schedule for regular maintenance and inspection of the earthing system and protective devices as per BS 7671 Regulation 651.2. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on proper earthing practices and the importance of maintaining electrical safety standards in line with BS 7671 requirements.



Figure 2.4 - Earth connection was found missing in distribution board near main gate.



Figure 2.5 - Earth wire was found missing in cutting machine power plug at first floor.



Figure 2.6 - undersized earth continuity conductor was found in panel near main gate.



Figure 2.7 - Earth connection was found missing in grey area ground floor.



Figure 2.8 - The earth bar was found without any connected cables.



Figure 2.9 - Earth connection was found missing in power factor improvement panel.

6.4 Panel/cabling/busbar

This relates to the entire facility.

The visual site inspection included an examination of the electrical panels, cables and busbars throughout the facility.

i. In a number of external areas it was observed that electrical components were not suitable for the environmental conditions in which they were installed such as fire pump panel.
ii. Several electrical panels such as changeover panel at generator area, exhibited signs of corrosion, indicating potential exposure to moisture or other corrosive elements.
iii. There were visible signs of damage to panels and its doors near main gate.
iv. The neutral bar was found installed too close to the panel door, creating a risk of contact and posing a serious electric shock hazard.provide
The utilisation of electrical equipment that is either unsuitable for its operating environment or physically damaged poses significant safety risks. These risks include potential electrical failures, increased likelihood of short circuits, and the heightened risk of fire and electrical shock. F-202604-065435

Timeline	Remediation Plan
Medium Term CAP-0077711	i. Identify and replace electrical panels, cables, and busbars that are not suitable for the environmental conditions in which they are installed in accordance with Section 512.2 of BS 7671 18th Edition + A2 (2022).
Medium Term CAP-0077712	ii. Replace or refurbish corroded electrical panels to prevent further degradation and potential safety hazards. Implement protective measures such as installing weatherproof enclosures or using corrosion-resistant materials.
Medium Term CAP-0079019	iii. Inspect and repair or replace all damaged panels, cables, and busbars to ensure the integrity of the electrical system in accordance with Section 522 of BS7671 18th Edition + A2 (2022).
Medium Term CAP-0079020	iv. Reposition the neutral bar to maintain safe clearance from the panel door and ensure proper insulation/guarding to prevent accidental contact.
Longer Term CAP-0079021	Establish a schedule for regular maintenance and inspection of the panel, cabling and busbars. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on the importance of maintaining electrical safety standards in line with BS 7671 requirements.



Figure 2.10 - Broken panel door was found near main gate in the facility.



Figure 2.11 - The neutral bar was installed too close to the panel door.



Figure 2.12 - Deteriorated panel with rust and corrosion was found in the generator area.



Figure 2.13 - Fire pump panel was found in outdoor area without weather protection.

6.5 Loop Connection

This relates to the entire facility.

During the inspection loop connections were identified within the electrical panel such as Grey area at ground floor.

In these loop connections, one circuit breaker is energised by the live part of another circuit breaker, creating loops within the panel instead of using a busbar system.

Avoiding loop connections is crucial due to several significant risks. Loop connections can complicate maintenance activities, making it difficult to trace and isolate circuits. These connections can lead to increased resistance and overheating, potentially damaging components. Loop connections raise the risk of accidental phase mixing, which can result in short circuits. The above issues compromise personal safety and significantly increase the risk of fire.F-202604-065442

Timeline	Remediation Plan
Medium Term CAP-0077746	All electrical panels shall be modified to eliminate loop connections. A busbar system within the electrical panels shall be utilised to provide each circuit with the required live, neutral, and earth connections.



Figure 2.14 - Loop connection was found in grey area at ground floor.

6.6 Multiple Connections

This relates to the whole facility.

During our inspection, multilooping was observed in a number of electrical panels across the facility.

In these cases, multiple circuits were being protected by a single circuit breaker at different locations such as grey area at ground floor and third floor panel.

Multilooping can result in improper distribution of electrical current, leading to voltage drops and potential electrical hazards. To mitigate these risks and ensure the safety and reliability of the electrical system, it is essential to adhere to proper installation practices. This includes using appropriate dedicated circuit breakers for each circuit.F-202604-065438

Timeline	Remediation Plan
Medium Term CAP-0077720	The electrical installation throughout the facility shall be modified to ensure each circuit is protected by a dedicated circuit breaker of a suitable type and rating for each individual circuit in accordance with Section 430 of BS 7671 18th Edition + A2 (2022)



Figure 2.15 - Multiple connection was found in grey area.

6.7 Openings

This relates to the entire facility of the factory.

During the visual inspection, the electrical panels and distribution boards were examined for unsealed openings to ensure compliance with the IP rating requirements.

Unsealed openings were discovered in the panels and distribution boards

These deficiencies pose multiple hazards. Unsealed openings and missing panels expose live electrical components, significantly increasing the risk of accidental contact and electric shock to personnel working nearby. The unsealed openings allow dust, debris, and moisture to enter, which can lead to short circuits and increase the risk of fire. The absence of proper sealing compromises the IP rating of the panels, which is crucial for ensuring safety and protection against environmental factors.F-202604-065453

Timeline	Remediation Plan
Medium Term CAP-0077921	Corrective actions are required to seal all openings and replace missing panels to restore the integrity and safety of the electrical systems. The electrical system shall ensure compliance with Sections 614.2.1, 412.2.2, and 512.2 of BS 7671 18th Edition + A2 (2022).
Longer Term CAP-0077922	Conduct regular maintenance and maintain comprehensive records for all electrical panels to ensure there are no openings and that the IP ratings of the panels remain uncompromised.



Figure 2.16 - openings were found in changeover panel.



Figure 2.17 - Opening was found in main panel near main gate.

6.8 Adequate accessibility

This relates to the entire facility.

During the visit, the electrical panels and distribution boards were examined for adequate accessibility.

At a few locations, such as the lift panel room, it was observed that the room was not accessible due to missing stairs or any other appropriate access arrangement to reach the room and the panel inside.

Limited accessibility to distribution panels and machine panels hampers effective maintenance and inspection procedures. This can potentially cause delays in addressing emergencies and troubleshooting electrical problems. These accessibility challenges could compromise the operational reliability and safety of the premises, as prompt maintenance and emergency responses are hindered. F-202604-065471

Timeline	Remediation Plan
Longer Term CAP-0077934	Provide permanent and safe access to lift panel rooms by installing fixed stairs, ladders, or platforms as per applicable safety standards.



Figure 2.18 - Access to the lift panel room was found to be missing.

6.9 Lint & dust inside panel board

This relates to the entire facility.

During the inspection, the cleanliness of electrical panels, distribution boards such as main panel near main gate, changeover panel and electrical containment was assessed.

The inspection revealed a significant accumulation of lint and dust in most of the distribution boards, panels.

The presence of lint and dust inside distribution boards poses several hazards. Accumulated lint and dust can act as fuel for fires if exposed to electrical currents or sparks, significantly increasing the risk of electrical fires. The cooling capacity of electrical components can also be impaired by the presence of dust and lint, leading to overheating.

Additionally, the accumulation of dust and lint can obstruct proper operation and maintenance, thereby impairing the functionality of the electrical components.F-202604-065454

Timeline	Remediation Plan
Medium Term CAP-0077923	All electrical panels and trunking runs shall be inspected and cleaned of any dust or lint in accordance with Section 522.4.2 of BS7671 18th Edition + A2 (2022)
Longer Term CAP-0077924	A regular maintenance procedure shall be established to ensure routine inspection and cleaning of all electrical panels and trunking runs. This will prevent the accumulation of lint and dust, thereby maintaining the safety and efficiency of the electrical systems.



Figure 2.19 - Lint and dust was found in changeover panel.



Figure 2.20 - Lint and dust was found in main panel near main gate.



Figure 2.21 - Lint and dust was found in main panel near main gate.

6.10 Lightning Protection System (LPS)

During the inspection, the lightning protection strategy of the factory was assessed.

A lightning protection system was provided, however, there was no study available to confirm its adequacy.

Lightning strikes can cause significant damage to buildings, primarily through the ignition of fires. When lightning strikes a structure, it can cause an electrical surge that ignites flammable materials. Without a proper lightning protection system (LPS), the energy from a lightning strike can travel through the building's electrical and plumbing systems, creating sparks and starting fires. F-202604-065764

Timeline	Remediation Plan
Medium Term CAP-0078195	Engage a qualified lightning protection specialist to perform a comprehensive risk assessment. This assessment will determine the necessity of a lightning protection system (LPS) based on factors such as building height, location, occupancy, and the presence of sensitive electronic equipment.
Longer Term CAP-0078196	If the risk assessment concludes that an LPS is necessary, proceed with the design and installation of an appropriate system in compliance with relevant standards such as BS EN 62305 (Protection against Lightning).

7. Fire Alarm & Detection System (FADS)

7.1 Non-functional or incomplete

This section refers to the entire facility.

During the inspection, the Fire Alarm and Detection System (FADS) for the facility was evaluated.

A conventional fire detection and alarm system was installed in the facility. Moreover it was found to be non-listed, non-compliant, having inadequate detection coverage and non-listed equipment pose life safety risk in the facility.
A non-functional/incomplete FADS significantly increases the risk of worker injury or fatality, substantial property damage, operational disruptions, legal liabilities, reputational damage, and environmental harm.F-202604-065444

Timeline	Remediation Plan
Medium Term CAP-0077760	i. Submit design documentation for new FADS system to Pakistan Accord for review before purchasing any new equipment. Design documentation to include drawings, technical data sheets, and product certifications of all listed equipment to show full compliance with the Fire Alarm and Detection System (FADS) with NFPA 72.
Longer Term CAP-0077759	ii. Submit documentation to the Pakistan Accord showing that an L/C has been opened for import of any components required for the FADS design submitted earlier.
Longer Term CAP-0077761	iii. Submit documentation to the Pakistan Accord demonstrating that all materials, both imported and locally procured, have been purchased and are available on-site for installation.
Longer Term CAP-0077762	iv. Once installation of the FADS system is complete, the factory is to submit full "as-built" documentation of the system to the Pakistan Accord for review.
Longer Term CAP-0077763	v. Factory to advise Pakistan Accord when FADS installation is fully completed and ready for review. Pakistan Accord will then arrange a verification inspection to ensure that the installation fully complies with the previously reviewed design documentation.

Longer Term CAP-0077764	vi. Provide full testing and commissioning (T&C) certification from the FADS system installer and make provision for attendance of the system installer, factory technical personnel, and Pakistan Accord technical team to witness a final T&C demonstration verification on site. On successful completion of this T&C verification inspection, remediation of the FADS system will be complete.

8. Emergency Lighting

8.1 Provided but non-functional

The section refers to the entire facility.

During the inspection it was observed that emergency lighting was provided but not functional in a number of locations such as grey area and 5th floor.

Functional emergency lighting is critical for providing illumination in the event of a power failure or other emergencies, ensuring that workers can safely evacuate the facility.F-202604-065443

Timeline	Remediation Plan
Medium Term CAP-0077750	Conduct a thorough inspection of all emergency lighting units throughout the facility to identify specific locations and units that are non-functional. This involves checking both the lights and the power sources, including batteries and wiring connections. Repair any faulty emergency lighting units immediately. If repair is not feasible, replace the defective units with new, compliant emergency lighting fixtures to ensure they provide adequate illumination during an emergency. Dim emergency light was found at 4th floor.
Longer Term CAP-0077751	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.22 - Non-functional emergency light was found at 5th floor.



Figure 2.23 - Non-functional emergency light was found at ground floor.

8.2 Provided but insufficient

The section refers to the entire facility.

During the inspection the emergency lighting arrangement throughout the facility was assessed.

The following observations were made:

i. Several areas lack sufficient emergency lighting coverage such as at second floor.
ii. Installed emergency lights are too dim to provide adequate illumination at ground and second floor.
iii. The backup power supply to the emergency luminaires is insufficient at different locations such as ground floor storage area, generator area.
iv. External emergency lighting was not provided, preventing people from reaching a place of safety outside the facility during night time evacuation.

Adequate and functional emergency lighting throughout the facility is crucial for providing necessary illumination during power failures or other emergencies, ensuring the safe evacuation of workers.F-202604-065445

Timeline	Remediation Plan
Medium Term CAP-0077906	i. Conduct a comprehensive survey to identify areas lacking adequate emergency lighting. Develop design drawings as per Building Code of Pakistan Fire safety provisions 2016 and submit to Pakistan Accord for its review and feedback. Install additional emergency luminaires in these areas to meet the coverage requirements specified in the Building Code of Pakistan Fire Safety Provisions 2016, ensuring sufficient illumination for safe evacuation
Medium Term CAP-0077907	ii. Replace dim emergency lights with higher intensity luminaires that meet the minimum illumination levels specified in the Building Code of Pakistan Fire Safety Provisions 2016. Ensure that all emergency lights provide adequate brightness to facilitate safe evacuation during emergencies
Medium Term CAP-0077908	iii. Upgrade the backup power supply to ensure it can adequately support all emergency luminaires during a power failure. This includes installing additional battery packs or generators as necessary to comply with the Building Code of Pakistan Fire Safety Provisions 2016.
Medium Term CAP-0077909	iv. Install external emergency luminaires to enable safe evacuation during hours of darkness all the way to a safe place outside the building(s)

Longer Term CAP-0077910	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.24 - Signage light was found to be dim in the facility.



Figure 2.25 - Signage light was found to be dim at second floor in the facility.



Figure 2.26 - Emergency lighting was found missing in staircase area.



Figure 2.28 - Insufficient coverage of emergency lighting was found in second floor stitching area.



Figure 2.27 - Emergency lighting was found with very low backup.



Figure 2.29 - Emergency lighting fixtures with zero backup was found.



Figure 2.30 - External emergency light was found missing in the facility.

9. Generator, Transformer & Sub-Station Room

9.1 Battery

During the inspection of the battery system, the battery cells, connections, and enclosures were thoroughly examined.

Several battery terminals were exposed and not properly insulated, increasing the risk of electrical shock or short circuits.

Proper monthly and annual maintenance of the battery system is crucial for fire safety. Spilled fluid can cause corrosion and fires, while bare terminals increase the risk of electrical shocks and short circuits, potentially igniting fires or explosions. Non-compliant battery placement can obstruct ventilation, leading to hazardous gas buildup and fire risks. Addressing these issues is essential to prevent fires and ensure battery system safety.F-202604-065762

Timeline	Remediation Plan
Medium Term CAP-0078191	Insulate exposed battery terminals using appropriate insulating materials to prevent electrical shock and short circuits. Conduct a thorough inspection of all connections to ensure that there are no immediate risks of exposure.
Longer Term CAP-0078192	Establish a routine inspection and maintenance protocol to regularly check and maintain proper insulation of all battery terminals.



Figure 2.31 - Battery exposed connections were observed.

10. Cabling & Wiring

10.1 Cable support and protection

This section refers to the entire facility.

During the inspection of the entire facility, a visual survey of the cable containment system was conducted.

i. Proper supports were not provided for cables; they were laid directly on the ground and hung along the walls. Additionally, several locations had open and exposed electrical connections. Inadequate cable support results in sagging or stress on cables, which can lead to damage or failure.

ii. Air pressure pipes were found running inside cable trays which need to be separated to avoid electrical and mechanical risks. F-202604-065447

Timeline	Remediation Plan
Medium Term CAP-0077919	i. Secure cables with appropriate supports and insulate any exposed connections to prevent sagging, stress, and potential hazards. (Reference: BS 7671 18th Edition + A2 (2022) Regulation 522.8.5 for mechanical support and 526.5 for proper terminations).
Longer Term CAP-0077920	ii. Air pressure pipes shall be removed from the cable trays to avoid any hazard.
Longer Term CAP-0079024	Implement a comprehensive cable management system with regular inspections to ensure cables remain properly supported and insulated over time.



Figure 2.32 - Inappropriate supporting arrangement for the cover of the cable tray.



Figure 2.34 - Inappropriate cable supports were provided.



Figure 2.33 - Clearance around cable trays have not been maintained.



Figure 2.35 - Inappropriate cable supports were provided in generator area.



Figure 2.36 - Air pressure pipes are running in a cable tray along with cables.



Figure 2.37 - Air pressure pipes are running in a cable tray along with cables.



Figure 2.38 - Cable supports were found missing at roof area.

10.2 Splicing of cables

This relates to the whole facility.

During the inspection the cable installation was visually surveyed throughout the facility.

i. Cables terminations within panels and switchgears were made without cable lugs, resulting in breaking strands and loose connections. This presents a significant fire risk due to the potential for electrical arcing and overheating, which can ignite nearby materials and cause a fire. Furthermore, loose connections can cause voltage drops and equipment malfunctions, posing a life safety hazard.
ii. PVC tape joints were found in several locations - including within distribution boards - where cables were either damaged or two cables were being joined together. This method of jointing is unreliable and can deteriorate over time, increasing the risk of electrical shorts and fires. The adhesive on PVC tape can degrade, leading to exposed conductors that can spark and ignite surrounding materials, presenting both fire and life safety risks.
iii. Cables with damaged insulation were found, with exposed conductors posing an electric shock risk within the facility.
For any future electrical work, the termination of stranded cables must be done with cable lugs and PVC tape joints shall not be utilised. Proper termination ensures secure and reliable connections, reducing the risk of electrical failures and enhancing overall safety. F-202604-065451

Timeline	Remediation Plan
Medium Term CAP-0077690	i. install cable lugs on all improperly terminated cables within panels and switchgears to secure connections and prevent breaking strands and loose connections in accordance with Regulation 526.1 of BS7671 18th Edition + A2 (2022)
Medium Term CAP-0077691	ii. Replace all PVC tape joints with proper terminal blocks or connectors to ensure secure and stable connections in accordance with Regulation 526.1 of BS7671 18th Edition + A2 (2022). Replace all damaged cabling with new wiring.
Medium Term CAP-0078185	iii. All cables with damaged insulation shall be replaced or properly repaired using appropriate methods. Protect cables using appropriate containment and maintain insulation integrity to eliminate electric shock risk.
Longer Term CAP-0078186	i. Implement a standard operating procedure (SOP) requiring the use of cable lugs for all future cable terminations. Regularly train and audit personnel to ensure compliance with this procedure, thus maintaining safe and reliable electrical connections.

Longer Term CAP-0078187	ii. Establish a policy prohibiting the use of PVC tape for electrical jointing. Regularly inspect and maintain electrical installations to ensure compliance, and provide continuous training for staff on proper jointing techniques.
Longer Term CAP-0079364	iii. Implement and enforce a cable integrity policy mandating the use of properly insulated cables, immediate replacement of damaged insulation (no temporary repairs), periodic inspections, and regular staff training to ensure ongoing electrical safety compliance.



Figure 2.39 - Cable was found with damaged insulation in the facility.



Figure 2.40 - Cable terminations without lugs at third floor.



Figure 2.41 - PVC joint was found mezzanine floor panel.



Figure 2.42 - PVC tape joint was found in mezzanine panel.



Figure 2.43 - Cable terminations were found without cable lugs at grey area ground floor.

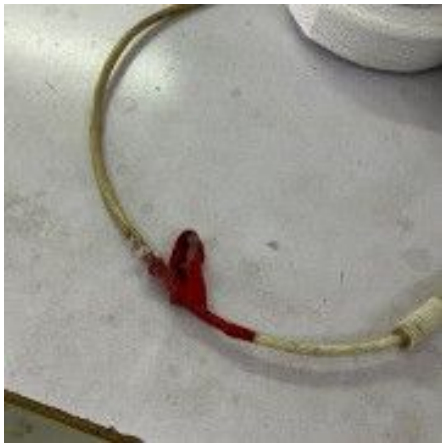


Figure 2.44 - PVC tape joints were found with machine cable at 4th floor.

10.3 Colour coding

This relates to the whole facility at different locations such as LT panel and water cooler DB at roof top.

During the inspection of several distribution boards, deviations from standard cable colour coding practices were identified. Such deviations pose significant risks, as they can lead to the misinterpretation and misidentification of cables. This misidentification can result in incorrect connections, potentially causing unsafe working conditions and increasing the likelihood of electrical accidents, such as shocks or fires.

To mitigate these risks, it is imperative that all future electrical work strictly adheres to standardized colour coding as prescribed by BS 7671. This will ensure accurate identification of conductors, thereby enhancing overall safety and operational efficiency. F-202604-065450

Timeline	Remediation Plan
Medium Term CAP-0077528	Conduct a comprehensive audit of all distribution boards to identify and correct any deviations from the standard cable colour coding. This involves re-labelling or re-insulating the cables with the appropriate colour codes as specified in BS 7671. Specifically, adhere to BS 7671 Regulation 514.3.1, which mandates the correct identification of conductors.
Longer Term CAP-0077529	Envelop and enforce a strict protocol for cable colour coding to be followed for all future installations and maintenance activities. This protocol should include regular training sessions for electricians and maintenance personnel to keep them updated on the latest standards and practices. Additionally, periodic audits should be conducted to ensure ongoing compliance with the standardized colour coding system, thereby preventing future deviations and ensuring sustained safety and operational efficiency.



Figure 2.45 - Color code deviations were found in water cooler DB at rooftop



Figure 2.46 - Color code deviations were found in LT panel.

10.4 Fixtures

This relates to the whole facility.

During the inspection, the electrical fixtures throughout the facility were visually surveyed.

i. Power receptacles mounted on combustible wooden ply: Installing receptacles on combustible materials like wooden ply increases the risk of fire if a short circuit or overheating occurs. Combustible materials can easily ignite, leading to potential fire hazards and endangering life safety. It was observed prayer area, first floor and 4th floor.
ii. Cables are routed along wooden table structure without proper containment.F-202604-065436

Timeline	Remediation Plan
Medium Term CAP-0077716	i. Replace combustible wooden ply backings with non-combustible materials to reduce fire risk.
Medium Term CAP-0077717	ii. All electrical accessories mounted on combustible wooden structures shall be relocated to approved non-combustible enclosures.
Longer Term CAP-0078188	Establish a facility-wide standard to use only non-combustible materials for mounting electrical components, and conduct regular audits for compliance.



Figure 2.47 - Wooden switchboard was found in cutting area at first floor.



Figure 2.48 - Wooden board was found in first floor area.

11. Assessment Limitations and Assumptions

The information in this report was obtained during an inspection of the facility and based on information provided by factory management. It has not been possible to provide independent verification for all the information and data collected, and therefore, the Accord cannot accept general responsibility for omissions or errors arising from inaccuracies in this report from the information obtained.

The findings and recommendations in this report are not intended to imply, guarantee, ensure or warrant compliance with any government regulations. Additionally, the results do not imply in any way that compliance with the findings or recommendations as stated in this report will eliminate all hazards, risks or exposures or that hazards, risks or exposures not referred to in this report do not exist. Compliance with the findings and recommendations stated in this report does not relieve the factory owner from obligation to comply with specific project requirements, industry standards, or the provisions of any local government regulations.

Except as otherwise noted within this report, other systems or services were not viewed or tested during the assessment and are therefore similarly excluded from this report. We have not assessed any parts of the structure / building fabric / services which are covered, unexposed or inaccessible and we are therefore unable to report that any such part of the property is free from defect.

The observations in this report are based on the engineering judgement of the engineer at the time of the inspection.